

Problem

The voltage across a device is given by

$$v(t) = -2 + 10 \cos 4t + 8 \cos 6t + 6 \cos 8t \\ - 5 \sin 4t - 3 \sin 6t - \sin 8t \text{ V}$$

Find:

(a) the period of $v(t)$,

(b) the average value of $v(t)$,

(c) the effective value of $v(t)$.

Step-by-step solution

Step 1 of 3

(A) For the first two AC terms, the frequency ratio is $\frac{6}{4} = 1.5$ so that the highest commonest is 2.

Hence $\omega_o = 2$

$$T = \frac{2\pi}{\omega_o} = \frac{2\pi}{2} = \pi$$

Step 2 of 3

(B) The average value is the DC component = - 2

Step 3 of 3

$$(C) \quad V_{rms} = a_o + \frac{1}{2} \sum_{n=1} (a_n^2 + b_n^2)$$

$$V_{rms}^2 = (-2)^2 + \frac{1}{2} (10^2 + 8^2 + 6^2 + 3^2 + 1^2) = 121.5$$

$$V_{rms} = 11.02 \text{ V}$$